OBINNA DOMINIC UKE

Applied Environmental Economist ◇ Climate, Energy, and Finance

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Africa Business School, Mohammed VI Polytechnic University (UM6P), Rabat, Morocco
COPPE, Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil

FIELDS

Primary: Environmental and Energy Economics
Secondary: Financial Economics, Development Economics, Applied Econometrics

Research interests: Carbon pricing and fossil fuel subsidies; fossil revenue, carbon lock-in and transition risk; carbon border adjustment and carbon accounting; climate risk and financial markets; sovereign risk and credit markets; climate finance and adaptation in the Global South.

EDUCATION

Ph.D. in Management and Organizational Sciences 2023 to Aug. 2027 (expected)

Research field: Environmental and Energy Economics
Africa Business School, Mohammed VI Polytechnic University (UM6P), Rabat, Morocco

- Dissertation: *Climate Risk, Carbon Pricing, and Financial Markets: Essays in Environmental and Financial Economics*
- Supervisors: Prof. Amine Ben Amar (Finance Cluster, ABS, UM6P) and Prof. Olga Gorelkina (Moroccan Center for Game Theory, UM6P)

Joint Doctorate, Production Engineering (PEP/COPPE) 2023 to Aug. 2027 (expected)

Research field: Environmental and Energy Economics
COPPE, Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil

- Joint doctoral agreement with ABS, UM6P, conferring the doctoral degrees of both institutions on a single defence; co-supervisor Prof. Luan Santos

M.Sc. in Fertilizer Science and Technology Oct. 2020 to Jul. 2022
ESAFE, Mohammed VI Polytechnic University (UM6P), Benguerir, Morocco

B.Sc. in Agriculture (Soil and Crop Sciences) Dec. 2013 to Aug. 2018
Usmanu Danfodiyo University, Sokoto, Nigeria

JOB MARKET PAPER

"Where the Windfall Goes: Fossil Revenue, Domestic Absorption and Carbon Emissions"

Sole authored.

Using a windfall shock for 163 countries over 1981 to 2024, built from predetermined fossil rents and unexpected world fuel prices and instrumented with OPEC oil supply news, I show that fossil windfalls reach the government budget, are saved abroad in the first year and spent within two. Each dollar of windfall revenue adds 0.52 kg of CO_2 at home (weak-instrument-robust interval 0.20 to 0.89), about half the carbon intensity of an average dollar of GDP, mostly through road transport, power and buildings. In cement, buildings and power, booms raise emissions that busts do not reverse. Over the 2003 to 2014 boom, windfalls added about 2.7 billion tonnes of CO_2.

WORKING PAPERS

"Riding the Climate Wave: A Wavelet Analysis of Sovereign Risk in Developed and Developing Economies" (with Amine Ben Amar and Olga Gorelkina). *Revise and resubmit, International*

Review of Financial Analysis; revision submitted September 2026.

"Fiscal Capacity, Not Climate Exposure: Decomposing Global Fossil-Fuel Underpricing and Its Equity Burden" (with Olga Gorelkina and Amine Ben Amar). *Submitted, Energy Policy.*

"Financial Reach and Household Coping under Climate Shocks: Evidence from Ten African Countries" (with Todd Cort). *Submitted, World Development.*

"Fossil Fuel Underpricing, Climate Disasters and the Carbon Composition of Manufacturing: Evidence from Global Value Chains". *Submitted, Economic Analysis and Policy.*

"Heat beyond the Farm Gate: Offsetting Supply and Demand Responses in European Food and Drink Manufacturing" (with Amine Ben Amar and Olga Gorelkina), 2026.

WORK IN PROGRESS

"Carbon Accounting Capability and Exposure to the EU Carbon Border Adjustment Mechanism: Evidence from African Exporters" (with Luan Santos). Target: Business Strategy and the Environment.

"Climate Shocks and Food Manufacturing: Aggregation Bias in the United Kingdom" (with Amine Ben Amar and Olga Gorelkina).

"Sovereign Climate Risk and Corporate Bond Spreads: Evidence from Emerging Markets" (with Luan Santos).

"Building Resilience, Reducing Risk: A Systematic Review of Climate Resilience, Climate Finance Impacts, and Market Mispricing in the Global South" (with Amine Ben Amar and Olga Gorelkina).

PEER-REVIEWED PUBLICATIONS

Guéablé, Y.K.D., **Uke, O.D.** et al. (2022). Study of the Sodicity of Phosphate By-Products and Sludge Mixture for Large-Scale Application in Mine Site Reclamation. *Environmental Sciences Proceedings*, 16(1), 18.

Uke, O.D. & Haliru, M. (2021). Salinity Study of the Soils of Fadama Farms, Sokoto, Nigeria. *Farming and Management*, 6, 1-7.

Yusuf, Y.M., Bagudo, H.A. & **Uke, O.D.** (2021). Effects of Weed Control on Growth and Yield of Pearl Millet in the Sudan Savanna, Sokoto State, Nigeria. *Journal of Current Opinion in Crop Science*, 2(2), 155-164.

CONFERENCE PRESENTATIONS

3rd Quantitative Finance and Data Analytics Meeting (QFDAM 2026), Rabat, Morocco June 2026
Climate Shocks, Sectoral Transmission, and Monetary Policy: Evidence from UK Food Manufacturing

2nd Quantitative Finance and Data Analytics Meeting (QFDAM 2025), Amiens, France June 2025
Riding the Climate Wave: Wavelet Analysis of Sovereign Risk in Developed and Developing Economies

71st EURO Working Group for Commodities and Financial Modelling (EWGCFM 2025), Rabat, Morocco May 2025
Riding the Climate Wave: Wavelet Analysis of Sovereign Risk in Developed and Developing Economies

TEACHING

Teaching Assistant, Financial Econometrics [Master 2] Sep. to Nov. 2026
Mohammed VI Polytechnic University (UM6P) (Instructor: Prof. Amine Ben Amar)

Co-developed and taught Lecture 5 (structural VARs, Granger causality, local projections, time-varying local projections with stochastic volatility, wavelets) as two three-hour sessions, with a reproducible R student pack of nine scripts on bundled data and Exercise 8.3; reviewed submitted exercises from 21 students.

Teaching Assistant, Microeconomics [Licence 1] Sep. to Nov. 2026
Mohammed VI Polytechnic University (UM6P) (Instructor: Prof. Amine Ben Amar)

Teaching Assistant, Essentials of Financial Markets [M1] Jan. to Feb. 2025
Africa Business School, UM6P (Instructor: Prof. Amine Ben Amar)

PROFESSIONAL EXPERIENCE

Regional Field Project Manager 2022 to 2023
OCP Africa Nigeria, USAID/TradeHub partnership

Research Agronomist 2021 to 2022
OCP Africa Nigeria

Founder and Chief Technical Officer 2020 to 2022
AgriBox SARL, Morocco

Project Manager, Microfinance Operations 2019 to 2020
Bincom ICT Solutions (Social Lender), Nigeria

FELLOWSHIPS AND AWARDS

Full Doctoral Merit Scholarship, Africa Business School, UM6P, Morocco 2023

HULT Prize in Innovation, Mohammed VI Polytechnic University 2021

NARSAD Deliberative Polling Award, UM6P and Stanford University 2021

IBN Rochid Foundation Scholarship, M.Sc. Studies, OCP Group 2020

SKILLS AND LANGUAGES

Methods: shift-share designs and local projections with external instruments; panel fixed effects with weak-instrument-robust and randomisation inference; SVAR; wavelet coherence; within-household designs.

Datasets constructed: fossil windfall and sector emissions panel (163 countries, 1981 to 2024); effective carbon prices (174 countries); CBAM liability by tariff line (1,760 codes); harmonised household shock panels (ten African countries). Replication packages rebuild results from raw data.

Software: R, Stata, Python, SQL, GIS, Git. **Languages:** English, Igbo (native); Portuguese (intermediate); French (basic).